Definición 2.7 (Cardinalidad). $|A| = \#$ de elementos *distintos* de A.

- $|A \cup B| = |A| + |B| - |A \cap B|$
- $|A - B| = |A| - |A \cap B|$
- $|A \times B| = |A| \cdot |B|$
- $|P(A)| = 2^{|A|}$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Definición 2.6 (Operaciones). Definición de algunas operaciones:

- $A \cup B = \{x \mid x \in A \vee x \in B\}$.
- $A \cap B = \{x \mid x \in A \wedge x \in B\}$.
- $A - B = \{x \mid x \in A \wedge x \notin B\}$.
- $\overline{A} = \mathcal{U} - A$.
- $A \triangle B = A \cup B - A \cap B$.
- $A \times B = \{(x, y) \mid x \in A \wedge y \in B\}$.
- $P(A) = 2^A = \{X \mid X \subseteq A\}$
- $\neg(\exists x[P(A)]) = \forall x[\neg P(A)]$

$A = \{2,3,4\}$ $\quad B = \{2,5,6\}$

$A \cup B = \{2,3,4,5,6\}$ $\quad A \triangle B = \{3,4,5,6\}$

$A \cap B = \{2\}$ $\quad A - B = \{3,4\}$

$\mathcal{U} = \{2,3,4,5\}$ $\quad \overline{A} = \mathcal{U} - A$

$= \{5\}$

$A = \{0,1,3\} \rightarrow 2^3 = 8$

$P(A) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{0,1\}, \{0,2\}, \{0,3\}, \{0,1,3\}\}$

2.2. [*] Si $A = \{0, 2, 3\}$, $B = \{0, 1, 3\}$ y $C = \{1, 3, 4\}$, con $\mathcal{U} = \{0, 1, 2, 3, 4, 5, 6\}$, calcule:

(a) $[(B \triangle C) \times A] \cap [\overline{B} \times (A \cap C)]$

Definición 2.6 (Operaciones). Definición de algunas operaciones:

- $A \cup B = \{x \mid x \in A \vee x \in B\}$.
- $A \cap B = \{x \mid x \in A \wedge x \in B\}$.
- $A - B = \{x \mid x \in A \wedge x \notin B\}$.
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Definición 2.7 (Cardinalidad). $|A| = \#$ de elementos *distintos* de A.

- $|A \cup B| = |A| + |B| - |A \cap B|$
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- $|A \times B| = |A| \cdot |B|$
- $|P(A)| = 2^{|A|}$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

$B \cup C = \{0, \cancel{1,3}, 4\} \quad - \quad B \cup C = \{1,3\}$

$B \triangle C = \{0,4\}$

$B \triangle C = \{0,4\} \times A = \{0,2,3\}$

$B \triangle C \times A = \{\{0,0\}, \{0,2\}, \{0,3\}, \{4,0\}, \{4,2\}, \{4,3\}\}$

2.2. [*] Si $A = \{0, 2, 3\}$, $B = \{0, 1, 3\}$ y $C = \{1, 3, 4\}$, con $\mathcal{U} = \{\cancel{0}, \cancel{1}, 2, \cancel{3}, 4, 5, 6\}$, calcule:

(a) $[(B \triangle C) \times A] \cap [\overline{B} \times (A \cap C)]$

$B \Delta C \times A = \{\{0,0\}, \{0,2\}, \{0,3\}, \{4,0\}, \{4,2\}, \{4,3\}\}$

$\overline{B} = U - B \rightarrow \{2, 4, 5, 6\} \times A \cap C = \{3\}$

$\overline{B} \times (A \cap C) = \{\{2,3\}, \{4,3\}, \{5,3\}, \{6,3\}\}$

$[(B \Delta C) \times A] \cap [\overline{B} \times (A \cap C)] = \boxed{\{4,3\}}$

$\{\{0,0\}, \{0,2\}, \{0,3\}, \{4,0\}, \{4,2\}, \{4,3\}\} \cap \{\{2,3\}, \{4,3\}, \{5,3\}, \{6,3\}\}$

2.2. [*] Si $A = \{0, 2, 3\}$, $B = \{0, 1, 3\}$ y $C = \{1, 3, 4\}$, con $\mathcal{U} = \{0, 1, 2, 3, 4, 5, 6\}$, calcule:

(b) $P[(C - A) \cap (B - A)]$

$C - A = \{1, 4\} \cap B - A = \{1\}$

$P(\{1\}) \rightarrow 2^1$

$P[(C - A) \cap (B - A)] = \boxed{\{\emptyset, \{1\}\}}$

Definición 2.7 (Cardinalidad). $|A| = \#$ de elementos *distintos* de A.

- $|A \cup B| = |A| + |B| - |A \cap B|$
- $|A - B| = |A| - |A \cap B|$
- $|A \times B| = |A| \cdot |B|$
- $|P(A)| = 2^{|A|}$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

2.21. [*] Si $|A \cup B| = 5$, $|A| = 4$, $|B| = 2$, $|A - D| = 2$ y $|P(D)| = 8$, calule la cardinalidad de $P(A \cap B) \times (A \cup D) \times (A \cup B)$.

$\rightarrow 2^3 = 8$

$|A \cup B| = 5$

$|A| + |B| - |A \cap B| = 5$

$4 + 2 - |A \cap B| = 5$

$4 + 2 - 5 = |A \cap B|$

$|A \cap B| = 1$

$|A - D| = 2$

$|A| - |A \cap D| = 2$

$4 - |A \cap D| = 2$

$|A \cap D| = 2$

$D = 3$

$|A \cup D| = |A| + |D| - |A \cap D|$

$|A \cup D| = 4 + 3 - 2$

$|A \cup D| = 5$

$P(A \cap B) \times (A \cup D) \times (A \cup B)$.

$|A \cap B| = 1$ $\quad |A \cup D| = 5$ $\quad |A \cup B| = 5$

$P(1) = 2^1 = 2$

$2 \cdot 5 \cdot 5 = \boxed{20}$

Definición 2.7 (Cardinalidad). $|A| = \#$ de elementos *distintos* de A. $A \cap B = B \cap A$

- $|A \cup B| = |A| + |B| - |A \cap B|$
- $|A - B| = |A| - |A \cap B|$
- $|A \times B| = |A| \cdot |B|$
- $|P(A)| = 2^{|A|}$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

2.24. [*] Si $|A| = 6$, $|A \cap B| = 3$, $|P(B)| = 16$ y $|A - C| = 1$, determine:

(a) $|(B - A) \times (A \cap C)|$;

$|B - A| = |B| - |B \cap A|$

$|B - A| = 4 - 3$

$|B - A| = 1$

$P(B) = 16$

$2^4 = B = 4$

$|A - C| = 1$

$|A| - |A \cap C| = 1$

$6 - |A \cap C| = 1$

$|A \cap C| = 5$

$|(B - A) \times (A \cap C)|$

$1 \cdot 5$

$\boxed{5}$

2.24. [*] Si $|A| = 6$, $|A \cap B| = 3$, $|P(B)| = 16$ y $|A - C| = 1$, determine:

(b) $|P(P(A) \times (A \cap B))|$.

$P(6) = 2^6$

$= 64$

$A \cap B = 3$

$P(64 \times 3) = P(192) = \boxed{2^{192}}$

$$A \triangle B = A \cup B - A \cap B.$$

Definición 2.7 (Cardinalidad). $|A| = \#$ de elementos *distintos* de A.

- $|A \cup B| = |A| + |B| - |A \cap B|$ ▪ $|A - B| = |A| - |A \cap B|$ ▪ $|A \times B| = |A| \cdot |B|$ ▪ $|P(A)| = 2^{|A|}$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

2.26. Sean A y B conjuntos arbitrarios tales que $|A| = 3$, $|A \cap B| = 1$ y $|P(B)| = 32$. Determine:

(a) $|P(A \triangle B)| = |A \cup B| - |A \cap B| = 7 - 1 = \boxed{6}$

$|A| = 3$

$P(B) = 32 \to 2^5 = 32$ $\qquad |A \cup B| = |A| + |B| - |A \cap B|$

$B = 5$ $\qquad = 3 + 5 - 1$

$= \boxed{7}$

(b) $|B - A|$ $(|B| - |B \cap A|)$

$5 - 1$

$\boxed{4}$

Definición 2.7 (Cardinalidad). $|A| = \#$ de elementos *distintos* de A.

- $|A \cup B| = |A| + |B| - |A \cap B|$ ▪ $|A - B| = |A| - |A \cap B|$ ▪ $|A \times B| = |A| \cdot |B|$ ▪ $|P(A)| = 2^{|A|}$
- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

2.28. Dados A, B y C tales que $|A| = 5$, $|B| = 4$, $|A \cap B| = 2$, $|P(C)| = 8$ y $|C - A| = 1$. Determine:

(a) $|(A - B) \times P(A \cup C)|$

$|C - A| = 1$ $\qquad P(C) = 8 \to \boxed{C = 3}$

$|C| - |C \cap A| = 1$

$3 - |C \cap A| = 1$ $\qquad |A - B| = |A| - |A \cap B|$

$\boxed{|C \cap A| = 2}$ $\qquad = 5 - 2$

$\boxed{|A - B| = 3}$

$|A \cup C| = |A| + |C| - |A \cap C|$

$5 + 3 - 2$

$\boxed{|A \cup C| = 6}$

$P(6) = 2^6 = 64$ $\qquad (A - B| \times P(A \cup C)$

$3 \cdot 64$

$\boxed{192}$

c) $\forall x \in \mathbb{R} \; \exists y \in \mathbb{R} \; [x^2 = y]$ Verdadero

d) $\forall x \in \mathbb{R} \; \exists y \in \mathbb{R} \; [y^2 = x]$ Falso